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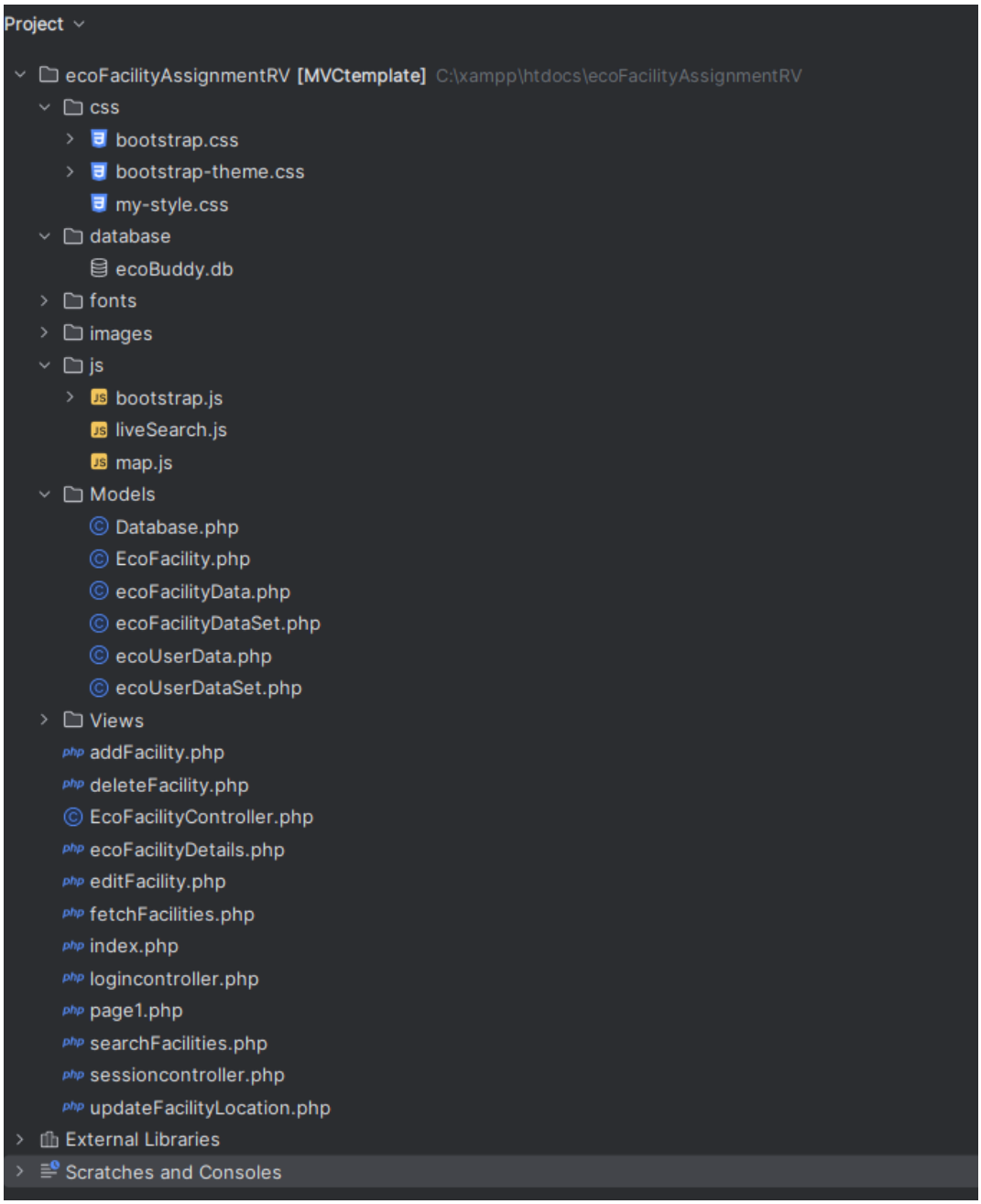
Title: Client Server systems Assignment 2

Project File Structure

http://localhost:63342/MVCtemplate/

Username - lee Password - H7k@M2!jXwY9r&Lp

Username – admin Password G7!fY8mZ$2k@PwR4



Mark range %	100-80% Outstanding to Excellent *(60-48 marks)*	79-60% Very Good to Good *(47-36 marks)*	59-40% Fair to Adequate *(35-24 marks)*	39-20% Unsatisfactory to Poor *(23-12 marks)*	19-0% Very p Extremely poo *(11-0 marks)*
	"An excellent system ready to deliver to the customer"	"A system with useful features"	"A minimum viable system with basic features"	"Incomplete features"	"Non- funct system"
Mapping **required feature** 60 marks	**Industry ready** application with **excellent performance** and **efficient data usage**. • Excellent OO code structure including reusable **classes**, design pattern(s) in both JavaScript and PHP as appropriate. *(up to 10 marks)* • Extensive use of **HTTPRequest** on **three or more** features of the solution UI to improve efficiency and performance using AJAX. *(up to 10 marks)* • Excellent and **secure input** validation, and demonstration of **security protection** such as URL tokens. *(up to 10 marks)* • **JSON or XML** data formats used for AJAX. PHP DB classes modified/extended to produce JSON/XML as necessary. *(up to 10 marks)* • **Sophisticated** map system with AJAX driven **real-time activity** and **updates**. Current user geo located. Outstanding user experience. Responsive layout. *(up to 10 marks)* • Excellent and consistently commented code. *(up to 10 marks)*	• Reusable JavaScript **functions** or **classes** added to perform robust input validation and displaying data to the users **dynamically**. • At least **two** AJAX type data transactions to acquire data from the PHP backend. • **JSON or XML** data formats used for AJAX. • **Useful** map system functioning using **AJAX**. Current user geo located. Very good user experience. Responsive layout • Good comments evident throughout.	• Some JavaScript added to perform input validation and/or displaying data to the users dynamically. • At least **one** AJAX type data transactions to acquire data from the PHP backend. • Plain text data format used for AJAX. • **Basic** map/list system functioning using AJAX. Acceptable user experience. • Some useful code comments evident. • Data is stored in an SQLIte file.	• Some JavaScript added to perform basic input validation and/or displaying location data to the users from PHP backend but non- functioning or incomplete/has issues and unsatisfactory system. Poor user experience. • **Significant amounts of copied code. Poorly presented document. Use of AI tools.** • Minimal code comments. • Use of prohibited tools such as jQuery, mySQL server.	• Little or no extra featu your seme • **Significant copied coo presented Use of AI t** • Little or no or experie • No code co • Use of pro such as jQ server.
Mark range %	**100-80%** Outstanding to Excellent *(40-32 marks)*	**79-60%** Very Good to Good *(31-24 marks)*	**59-40%** Fair to Adequate *(23-16)*	**39-20%** Unsatisfactory to Poor *(15-8 marks)*	**19-0%** Very p Extremely poo *(7-0 marks)*
Chosen feature 40 marks	**Industry ready** feature with **excellent performance** and **efficient data usage**. • **Excellent** OO code structure including reusable **classes**, design pattern, in both JavaScript and PHP as appropriate. *(up to 8 marks)* • Extensive use of **HTTPRequest** on different features of the feature to improve sophistication, efficiency and performance using AJAX. *(up to 8 marks)* • Excellent and **secure input** validation, and demonstration of **security protection** such as URL tokens. *(up to 8 marks)* • JSON or XML data formats used for AJAX. PHP DB classes modified/extended to produce JSON/XML as necessary *(up to 8 marks)* • Excellent and consistently commented code *(up to 8 marks)*	• Reusable JavaScript **functions** or classes added to perform robust input validation and displaying data to the users **dynamically**. • At least **two** AJAX type data transactions for your chosen feature. • JSON or XML data formats used for AJAX. • Good comments evident throughout.	• Some JavaScript added to perform robust input validation and/or displaying data to the users dynamically. • At least **one** AJAX type data transaction for the chosen extra feature. • Plain text data format used for AJAX. • Some useful code comments evident. • Data is stored in an SQLIte file.	• Some JavaScript added to perform basic input validation or displaying data to the users but non- functioning, incomplete and unsatisfactory extra feature. • **Significant amounts of copied code. Poorly presented document. Use of AI tools.** • Minimal code comments. • Use of prohibited tools such as jQuery, mySQL server.	• Little or no extra featu your seme • **Significant copied coo presented Use of AI t** • No code co • Use of pro such as jQ server.

Modified Models

EcoFacility.php

```php
<?php

no usages
class EcoFacility
{
    1 usage
    private $id;
    1 usage
    private $title;
    1 usage
    private $category;
    1 usage
    private $description;
    1 usage
    private $houseNumber;
    1 usage
    private $streetName;
    1 usage
    private $town;
    1 usage
    private $county;
    1 usage
    private $postcode;
    1 usage
    private $longitude;
    1 usage
    private $latitude;
    1 usage
    private $contributor;

    1 usage
    private $status;

    // Getter for ID
    no usages
    public function getID()
    {
        return $this->id;
    }

    // Getter for Title
```

```php
    public function getTitle()
    {
        return $this->title;
    }

    // Getter for Category
    no usages
    public function getCategory()
    {
        return $this->category;
    }

    // Getter for Description
    no usages
    public function getDescription()
    {
        return $this->description;
    }

    // Getter for Address components
    no usages
    public function getHouseNumber()
    {
        return $this->houseNumber;
    }

    //Getter for Street Name
    no usages
    public function getStreetName()
    {
        return $this->streetName;
    }

    //Getter for Town
    no usages
    public function getTown()
    {
        return $this->town;
    }
```

```php
    //Getter for County
    no usages
    public function getCounty()
    {
        return $this->county;
    }

    //Getter for Postcode
    no usages
    public function getPostcode()
    {
        return $this->postcode;
    }

    // Getter for Location components
    no usages
    public function getLongitude()
    {
        return $this->longitude;
    }

    //Getter for Latitude
    no usages
    public function getLatitude()
    {
        return $this->latitude;
    }

    // Getter for Contributor
    no usages
    public function getContributor()
    {
        return $this->contributor;
    }

    no usages
    public function getStatus(){
        return $this->status;
    }
```

```php

}
```

ecoFacilityData.php

```php
<?php
3 usages
class ecoFacilityData {

    // Define facility properties
    protected $id, $title, $category, $description, $houseNumber, $streetName, $county, $town, $postcode, $longitude, $latitude, $contributor, $status;
    1 usage
    private $pdo;

    //Constructor initializes properties from the database

    // binding user inputs into the variable to put into database
    3 usages
    public function __construct($dbRow){
        $this->id = $dbRow['id'];
        $this->title = $dbRow['title'];
        $this->category = $dbRow['category'];
        $this->description = $dbRow['description'];
        $this->houseNumber = $dbRow['houseNumber'];
        $this->streetName = $dbRow['streetName'];
        $this->county = $dbRow['county'];
        $this->town = $dbRow['town'];
        $this->postcode = $dbRow['postcode'];
        $this->longitude = $dbRow['lng'];
        $this->latitude = $dbRow['lat'];
        $this->contributor = $dbRow['contributor'];
        $this->status = $dbRow['status'];

    }

    // Method to insert a new facility into the database
    no usages
    public function insertNewEcoFacility($formData) {
        try {
            // Prepare the SQL query
            $stmt = $this->pdo->prepare(
                "INSERT INTO ecoFacilities (
                    title, category, description, houseNumber, streetName, county, town, postcode, lng, lat, contributor, status
                ) VALUES (
                    :title, :category, :description, :houseNumber, :streetName, :county, :town, :postcode, :lng, :lat, :contributor, :status
                )"
```

```
            );

            // binding user inputs into the variable to put into database
            $stmt->bindParam(':title', $formData['title']);
            $stmt->bindParam(':category', $formData['category']);
            $stmt->bindParam(':description', $formData['description']);
            $stmt->bindParam(':houseNumber', $formData['houseNumber']);
            $stmt->bindParam(':streetName', $formData['streetName']);
            $stmt->bindParam(':county', $formData['county']);
            $stmt->bindParam(':town', $formData['town']);
            $stmt->bindParam(':postcode', $formData['postcode']);
            $stmt->bindParam(':lng', $formData['lng']);
            $stmt->bindParam(':lat', $formData['lat']);
            $stmt->bindParam(':contributor', $formData['contributor']);
            $stmt->bindParam(':status', $formData['status']);

            // Execute the query
            $stmt->execute();

            return "Facility added successfully!";
        } catch (PDOException $e) {
            // If an error occurs, return the error message
            return "Error: " . $e->getMessage();
        }
    }
    //getter to return the ecoFacility ID
    3 usages
    public function getID() {
        return $this->id;
    }


    //getter returns ecoFacility title
    3 usages
    public function getTitle() {
        return $this->title;
    }

    //getter returns ecoFacility category
```

```php
    public function getCategory() {
        return $this->category;
    }

    //getter returns ecoFacility description
    2 usages
    public function getDescription() {
        return $this->description;
    }

    //getter returns ecoFacility house number
    2 usages
    public function getHouseNumber() {
        return $this->houseNumber;
    }

    //getter returns ecofacility street name
    2 usages
    public function getStreetName() {
        return $this->streetName;
    }


    //getter returns ecofacility county
    2 usages
    public function getCounty() {
        return $this->county;
    }

    //getter returns ecofacility town
    2 usages
    public function getTown() {
        return $this->town;
    }

    //getter returns ecofacility postcode
    2 usages
    public function getPostcode() {
        return $this->postcode;
    }
```

```php
109
110     //getter returns ecofacility longitude
        2 usages
111     public function getLongitude() {
112         return $this->longitude;
113     }
114
115     //getter returns ecofacility latitude
        2 usages
116     public function getLatitude() {
117         return $this->latitude;
118     }
119
120     //getter returns ecofacility contributor
        1 usage
121     public function getContributor() {
122         return $this->contributor;
123     }
124
125     //getter returns ecofacility status
        2 usages
126     public function getStatus(){
127         return $this->status;
128     }
129
130
131
132
133 }
```

ecoFacilityDataSet.php

```php
<?php
require_once('Database.php');
require_once('ecoFacilityData.php');

8 usages
class EcoFacilityDataSet {
    9 usages
    protected $_dbHandle;

    8 usages
    public function __construct() {
        $this->_dbHandle = Database::getInstance()->getdbConnection();
    }

    // Fetch all eco facilities with optional pagination
    2 usages
    public function fetchAllEcoFacilities($limit = null, $offset = null) {
        // Base query
        $sqlQuery = "SELECT * FROM ecoFacilities ORDER BY title";

        // Add pagination if limit/offset is provided
        if ($limit !== null && $offset !== null) {
            $sqlQuery .= " LIMIT :limit OFFSET :offset";
        }

        $stmt = $this->_dbHandle->prepare($sqlQuery);

        // Bind values for pagination
        if ($limit !== null && $offset !== null) {
            $stmt->bindValue( param: ':limit', (int)$limit, type: PDO::PARAM_INT);
            $stmt->bindValue( param: ':offset', (int)$offset, type: PDO::PARAM_INT);
        }

        $stmt->execute();

        $dataSet = [];
        while ($row = $stmt->fetch()) {
            $dataSet[] = new ecoFacilityData($row);
        }
```

```php
        return $dataSet;
    }

    // Fetch eco facilities with search and/or category filter, and pagination
    2 usages
    public function searchEcoFacilities($query = '', $category = '', $limit = 10, $offset = null) {
        $sqlQuery = "SELECT * FROM ecoFacilities WHERE 1"; // Base query (always true)

        $params = [];

        // Add filters dynamically
        if (!empty($query)) {
            $sqlQuery .= " AND (title LIKE :queryPartial)";
            $sqlQuery .= " ORDER BY CASE WHEN title LIKE :queryExact THEN 1 WHEN title LIKE :queryPartial THEN 2 ELSE 3 END";
            $params[':queryPartial'] = '%' . $query . '%'; // Matches anywhere in the title
            $params[':queryExact'] = $query . '%';         // Title starts with the letter

        }


        if (!empty($category)) {
            //filter results by the category
            $sqlQuery .= " AND category = :category";
            $params[':category'] = $category;
        }

        // Add pagination
        if ($limit !== null && $offset !== null) {
            $sqlQuery .= " LIMIT :limit OFFSET :offset";
        }


        $stmt = $this->_dbHandle->prepare($sqlQuery);

        // Bind dynamic parameters
        foreach ($params as $key => $value) {
            $stmt->bindValue($key, $value);
```

```php
        }

        // Bind pagination
        if ($limit !== null && $offset !== null) {
            $stmt->bindValue( param: ':limit', (int)$limit,  type: PDO::PARAM_INT);
            $stmt->bindValue( param: ':offset', (int)$offset,  type: PDO::PARAM_INT);
        }



        $stmt->execute();

        $dataSet = [];
        while ($row = $stmt->fetch()) {
            $dataSet[] = new ecoFacilityData($row);
        }
        return $dataSet;
    }

    // Total facility count for pagination
    2 usages
    public function getTotalRecords($query = '', $category = '') {
        $sqlQuery = "SELECT COUNT(*) AS total FROM ecoFacilities WHERE 1";

        $params = [];
        if (!empty($query)) {
            $sqlQuery .= " AND (title LIKE :query LIKE :query)";
            $params[':query'] = '%' . $query . '%';
        }

        if (!empty($category)) {
            $sqlQuery .= " AND category = :category";
            $params[':category'] = $category;
        }

        $stmt = $this->_dbHandle->prepare($sqlQuery);
        $stmt->execute($params);
```

```php
        $result = $stmt->fetch();
        return $result['total'];
    }

    1 usage
    public function deleteEcoFacility($id)
    {
        $sql = "DELETE FROM ecoFacilities WHERE id = :id";
        $statement = $this->_dbHandle->prepare($sql);
        //this deletes the ecofacility in the database where it's the id passed through

        try {
            // binding uthe statement into the variable to put into database
            $statement->bindParam( param: ':id', &var: $id, type: PDO::PARAM_INT);
            return $statement->execute(); // Returns true if success, if not, false
        } catch (PDOException $e) {
            // error log keeps track of potential errors
            error_log( message: "Deletion Error: " . $e->getMessage());
            return false;
        }
    }


    2 usages
    public function fetchEcoFacility($id) {
        $sqlQuery = "SELECT * FROM ecoFacilities WHERE id = :id";
        // Prepares the query to prevent SQL injection
        $stmt = $this->_dbHandle->prepare($sqlQuery);
        $stmt->execute(['id' => $id]);
        // the result should be an array
        $row = $stmt->fetch( mode: PDO::FETCH_ASSOC);
        //if the row is found it returns the ecofacility we searched for
        if ($row) {
            return new ecoFacilityData($row);
        }
        return null;
```

```
    }

    1 usage
    public function updateEcoFacility($id, $data) {
        //this is the sql query used to update an ecofacility and set the different attributes to what the user sets it as
        $sqlQuery = "UPDATE ecoFacilities
                    SET title = :title, category = :category, description = :description,
                        houseNumber = :houseNumber, streetName = :streetName, county = :county,
                        town = :town, postcode = :postcode, lng = :longitude, lat = :latitude, status = :status
                    WHERE id = :id";

        // Prepares the query to prevent SQL injection
        $stmt = $this->_dbHandle->prepare($sqlQuery);
        //executes the update, sets the different attributes in the database to what the user inputs
        $stmt->execute([
            // binding user inputs into the variable to put into database
            ':title' => $data['title'],
            ':category' => $data['category'],
            ':description' => $data['description'],
            ':houseNumber' => $data['houseNumber'],
            ':streetName' => $data['streetName'],
            ':county' => $data['county'],
            ':town' => $data['town'],
            ':postcode' => $data['postcode'],
            ':longitude' => $data['longitude'],
            ':latitude' => $data['latitude'],
            ':status' => $data['status'],
            ':id' => $id
        ]);
    }

    //Makes sure that facilities have unique titles
    2 usages
    public function isTitleUnique($title, $excludeId = null) {
        $sqlQuery = "SELECT COUNT(*) AS count FROM ecoFacilities WHERE LOWER(title) = LOWER(:title)";

        if ($excludeId !== null) {
```

```
            $sqlQuery .= " AND id != :id"; // Exclude the current record by ID
        }

        $stmt = $this->_dbHandle->prepare($sqlQuery);

        // Bind parameters
        $params = [':title' => $title];
        if ($excludeId !== null) {
            $params[':id'] = $excludeId;
        }

        $stmt->execute($params);
        $result = $stmt->fetch( mode: PDO::FETCH_ASSOC);

        return $result['count'] == 0; // Return true if the title is unique
    }

    1 usage
    public function addEcoFacility($formData) {
        $sql = "INSERT INTO ecoFacilities
                (title, category, description, houseNumber, streetName, county, town, postcode, lng, lat, contributor, status)
                VALUES
                (:title, :category, :description, :houseNumber, :streetName, :county, :town, :postcode, :lng, :lat, :contributor, :status)";

        $stmt = $this->_dbHandle->prepare($sql);

        $stmt->execute([
            // binding user input into the variables to put into database
            ':title' => $formData['title'],
            ':category' => $formData['category'],
            ':description' => $formData['description'],
            ':houseNumber' => $formData['houseNumber'],
            ':streetName' => $formData['streetName'],
            ':county' => $formData['county'],
            ':town' => $formData['town'],
            ':postcode' => $formData['postcode'],
            ':lng' => $formData['longitude'],
            ':lat' => $formData['latitude'],
            ':contributor' => $formData['contributor'],
```

```
            ':lat' => $formData['latitude'],
            ':contributor' => $formData['contributor'],
            ':status' => $formData['status']
        ]);
    }
```

Modified & New Controllers

deleteFacility.php

```php
<?php
require_once('sessioncontroller.php');
require_once('Models/EcoFacilityDataSet.php'); // Include the model to interact with eco facility data

if (!isset($_SESSION['userType']) || $_SESSION['userType'] != 1) {
    // Check if the user is logged in and is an admin (userType 1)
    die("Access Denied: You do not have permission to delete facilities.");
    // Stop the script if the user isn't an admin
}

if ($_SERVER['REQUEST_METHOD'] === 'POST' && isset($_POST['id'])) {
    // Check if the request is POST and the facility ID is provided in the request
    $id = intval($_POST['id']); // Ensure it's an integer
    $ecoFacilityDataSet = new EcoFacilityDataSet(); // Create an instance of the model to interact with

    if ($ecoFacilityDataSet->deleteEcoFacility($id)) {
        // Try deleting the facility with the given ID
        header( header: "Location: index.php?success=Facility deleted!"); // Redirect to the index page with a success message
        exit; // Stop execution after redirect
    } else {
        header( header: "Location: index.php?error=Failed to delete facility."); // Redirect to the index page with an error message
        exit; // Stop execution after redirect
    }
} else {
    die("Invalid Request: Facility ID not provided."); // Handle invalid requests
```

EcoFacilityController.php

```php
<?php
require_once('Models/EcoFacilityDataSet.php');

1 usage
class EcoFacilityController {
    1 usage
    public function viewAll() {
        require_once('sessioncontroller.php');

        $ecoFacilityDataSet = new EcoFacilityDataSet();

        // Pagination
        $facilitiesPerPage = 5;
        $currentPage = isset($_GET['page']) && is_numeric($_GET['page']) && $_GET['page'] > 0 ? $_GET['page'] : 1;
        $offset = ($currentPage - 1) * $facilitiesPerPage;

        // Search/filter logic
        $searchQuery = $_GET['query'] ?? '';
        $searchCategory = $_GET['category'] ?? '';

        // Fetch records and total count
        if (!empty($searchQuery) || !empty($searchCategory)) {
            //if the search query or category isn't empty, search for the ecoFacilities that meet the filter.
            $ecoFacilities = $ecoFacilityDataSet->searchEcoFacilities($searchQuery, $searchCategory, $facilitiesPerPage, $offset);
            $totalFacilities = $ecoFacilityDataSet->getTotalRecords($searchQuery, $searchCategory);
        } else {
            //show all ecoFacilities
            $ecoFacilities = $ecoFacilityDataSet->fetchAllEcoFacilities($facilitiesPerPage, $offset);
            $totalFacilities = $ecoFacilityDataSet->getTotalRecords();
        }

        $totalPages = ceil( num: $totalFacilities / $facilitiesPerPage);

        // Add properties to the view
        $view = new stdClass();
        $view->ecoFacilities = $ecoFacilities;
        $view->currentPage = $currentPage;
        $view->totalPages = $totalPages;
        $view->pageTitle = 'Eco Facilities';

        require_once('Views/index.phtml'); // Load the view file
```

```php
    }
}
```

ecoFacilityDetails.php

```php
<?php
require_once('Models/ecoFacilityDataSet.php');
require_once('sessioncontroller.php'); // Manage session timeout

// Get the facility ID from the URL
$id = $_GET['id'] ?? null;

if (!$id || !is_numeric($id)) {
    die("Invalid Facility ID!");
}

// Fetch the facility details using the EcoFacilityDataSet class
$dataSet = new EcoFacilityDataSet();
$facility = $dataSet->fetchEcoFacility($id);

if (!$facility) {
    die("Facility not found!");
}

// Include the view to display facility details
require('Views/ecoFacilityDetails.phtml');

// Include footer template
require('Views/template/footer.phtml');
?>
```

editFacility.php

```php
<?php

require_once('sessioncontroller.php');
require_once('Models/ecoFacilityDataSet.php');

// Check if the user is logged in and is an admin (userType 1)
if (!isset($_SESSION['userType']) || $_SESSION['userType'] != 1) {
    header( header: 'Location: index.php');
    exit;
}

// Check if the facility ID is provided in the URL
if (!isset($_GET['id'])) {
    header( header: 'Location: index.php'); // Redirect if no ID is provided
    exit;
}

$id = (int)$_GET['id']; // Get the ID from the URL

// Get the facility details
$ecoFacilityDataSet = new EcoFacilityDataSet();
$facility = $ecoFacilityDataSet->fetchEcoFacility($id);

// Handle form submission
if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $formData = [
        'id' => $id,
        'title' => $_POST['title'],
        'category' => $_POST['category'],
        'description' => $_POST['description'],
        'houseNumber' => $_POST['houseNumber'],
        'streetName' => $_POST['streetName'],
        'county' => $_POST['county'],
        'town' => $_POST['town'],
        'postcode' => $_POST['postcode'],
        'longitude' => $_POST['longitude'],
        'latitude' => $_POST['latitude'],
        'status' => $_POST['status']
    ];

    // Check for duplicate title (except the current record being edited)
    if (!$ecoFacilityDataSet->isTitleUnique($formData['title'], $id)) {
        // Title is not unique so display an error message
        $errorMessage = "Error: Another eco-facility with the same title exists. Please choose a different title.";
    } else {
        // Title is unique so proceed with update
        $ecoFacilityDataSet->updateEcoFacility($id, $formData); // Update method
        header( header: 'Location: index.php'); // Redirect on success
        exit;
    }
}

// Pass data to the view
$view = new stdClass();
$view->pageTitle = "Edit Eco-Facility";
$view->facility = $facility;
$view->errorMessage = isset($errorMessage) ? $errorMessage : null;

require_once('Views/editFacility.phtml');
?>
```

addFacility.php

```php
<?php

require_once('sessioncontroller.php');
require_once('Models/ecoFacilityDataSet.php');

// Check if the user is logged in and is an admin (userType 1)
if (!isset($_SESSION['userType']) || $_SESSION['userType'] != 1) {
    header( header: 'Location: index.php');
    exit;
}

// Handle form submission
if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $formData = [
        'title' => $_POST['title'],
        'category' => $_POST['category'],
        'description' => $_POST['description'],
        'houseNumber' => $_POST['houseNumber'],
        'streetName' => $_POST['streetName'],
        'county' => $_POST['county'],
        'town' => $_POST['town'],
        'postcode' => $_POST['postcode'],
        'longitude' => $_POST['longitude'],
        'latitude' => $_POST['latitude'],
        'contributor' => $_SESSION['login'],
        'status' => $_SESSION['status']
    ];

    // Create an instance of the EcoFacilityDataSet model
    $ecoFacilityDataSet = new EcoFacilityDataSet();

    // Check for duplicate title
    if (!$ecoFacilityDataSet->isTitleUnique($formData['title'])) {
        // Title is not unique; display an error message
        $errorMessage = "Error: Another eco-facility with the same title exists. Please choose a different title.";
    } else {
        // Title is unique
        $ecoFacilityDataSet->addEcoFacility($formData); // use the addEcoFacility method

        // Redirect to index page
        header( header: 'Location: index.php');
```

```php
            exit;
        }
    }

    // Pass page title and any error message to the view
    $view = new stdClass();
    $view->pageTitle = "Add Eco-Facility"; // Page is called eco-facility
    $view->errorMessage = isset($errorMessage) ? $errorMessage : null;

    require_once('Views/addFacility.phtml');
```

fetchFacilities.php

```php
<?php
require_once ('Models/EcoFacilityDataSet.php');

// Tell the browser that the output will be in JSON format
header( header: 'Content-Type: application/json');

// Create a new instance of EcoFacilityDataSet to access facility data
$ecoFacilityDataSet = new EcoFacilityDataSet();

// Get all eco facilities from the database
$facilities = $ecoFacilityDataSet->fetchAllEcoFacilities();

// Create an empty array to hold the formatted facility data
$response = [];

// Loop through each facility and get specific information
foreach ($facilities as $facility) {
    $response[] = [
        'id' => $facility->getID(),
        'name' => $facility->getTitle(),
        'latitude' => $facility->getLatitude(),
        'longitude' => $facility->getLongitude()
    ];
}

// Convert the response array to JSON format and print it
echo json_encode($response);
```

logincontroller.php

```php
<?php
session_start(); //start a session to manage users and data
require_once('sessioncontroller.php'); // Manage session timeout
require_once('Models/Database.php'); //  The database connection file

// Handle login
if ($_SERVER['REQUEST_METHOD'] === 'POST' && isset($_POST['username']) && isset($_POST['password'])) {
    $username = trim($_POST['username']);// Get the username from the login form.
    $password = trim($_POST['password']); // Get the password from the login form.

    $pdo = Database::getInstance()->getdbConnection();

    $stmt = $pdo->prepare( query: "SELECT * FROM ecoUser WHERE username = :username");
    $stmt->execute(['username' => $username]); // Run the query with the provided username.
    $user = $stmt->fetch( mode: PDO::FETCH_ASSOC); //It fetches the username

    if ($user && $password === $user['password']) {
        // Set session variables
        $_SESSION['login'] = $username;
        $_SESSION['userType'] = $user['userType'];  // Store usertype in session

        header( header: "Location: index.php"); // Redirect after successful login
        exit;
    } else {
        echo "Invalid username or password.";
    }
}

// check the logout button once pressed end session, log out
if ($_SERVER['REQUEST_METHOD'] === 'POST' && isset($_POST['logoutbutton'])) {
    session_destroy(); // Destroy the session
    header( header: "Location: page1.php"); // Redirect to the login page
    exit;
}
```

searchFacilities.php

```php
<?php

header( header: 'Content-Type: application/json'); // Set the response type to JSON, so the browser knows we are returning data
require_once('Models/Database.php');
require_once('Models/ecoFacilityDataSet.php');

//Get the query and category from the URL parameters, if not they're empty
$query = $_GET['query'] ?? '';
$category = $_GET['category'] ?? '';
$limit = 10;

// Make sure the search query is 1-30 characters long and only has letters numbers and spaces
if (!preg_match( pattern: '/^[a-zA-Z0-9\s]{1,30}$/', $query)) {
    //if not valid, return an empty JSON array and stop the script
    echo json_encode([]);
    exit;
}

//  Create an object
$dataSet = new EcoFacilityDataSet();

// Search for facilities based on the user's query and category
$facilities = $dataSet->searchEcoFacilities($query, $category);

// Format the data for the frontend to use
$jsonResults = array_map(function($facility) {
    return [
        'id' => htmlspecialchars($facility->getID()),
        'facilityName' => htmlspecialchars($facility->getTitle())
    ];
}, $facilities);

// Convert the PHP array into JSON and sent it back to the browser
echo json_encode($jsonResults);
?>
```

sessioncontroller.php

```php
<?php
// Check if a session has already started
if (session_status() === PHP_SESSION_NONE) {
    session_start(); // Start session only if no session is active
}

//  session timeout
define('SESSION_TIMEOUT', 900); // 900 seconds = 15 minutes

// Check if the user is logged in
if (isset($_SESSION['login'])) {
    //check if the last recorded activity time exists and if it's been more than 15 minutes
    if (isset($_SESSION['LAST_ACTIVITY']) && (time() - $_SESSION['LAST_ACTIVITY']) > SESSION_TIMEOUT) {
        // If it's been too long, inactivity, log them out
        session_unset();  // Unset session variables
        session_destroy(); // Destroy the session
        header( header: "Location: page1.php?timeout=true"); // Redirect to login page with timeout message
        exit;
    }
    // If the session hasn't timed out, update last activity time to now
    $_SESSION['LAST_ACTIVITY'] = time();
}
```

updateFacilityLocation.php

```php
<?php
require_once ('Models/EcoFacilityDataSet.php'); //EcoFacilityDataSet class
header( header: 'Content-Type: application/json'); // Set the response type to JSON

//Check if the request was made using POST method
if ($_SERVER['REQUEST_METHOD'] == 'POST') {
    // Get the facility ID and convert it to an integer
    $facilityID = intval($_POST['id']);
    $latitude = filter_var($_POST['latitude'], filter: FILTER_VALIDATE_FLOAT); // Validate latitude to make sure they are valid float numbers
    $longitude = filter_var($_POST['longitude'], filter: FILTER_VALIDATE_FLOAT); // Validate longitude to make sure they are valid float numbers

    //If any value is invalid, return an error message and stop
    if (!$facilityID || !$latitude || !$longitude) {
        echo json_encode(['success' => false, 'message' => 'Invalid input provided']);
        exit;
    }

    $ecoFacilityDataSet = new EcoFacilityDataSet();
    $result = $ecoFacilityDataSet->updateFacilityLocation($facilityID, $latitude, $longitude); // update the facility's location with new coordinates

    //Send back success status to frontend to confirm it worked
    echo json_encode(['success' => $result]);
}
```

Modified & New Views

addFacility.phtml

```php
<?php require('template/header.phtml'); ?>

    <div class="container mt-5">
        <!-- Page Heading -->
        <h2 class="text-center mb-4">Add a New Eco-Facility</h2>

        <!-- Display Error Message -->
        <?php if (isset($view->errorMessage) && !empty($view->errorMessage)): ?>
            <div class="alert alert-danger text-center">
                <?= htmlspecialchars($view->errorMessage, flags: ENT_QUOTES); ?>
            </div>
        <?php endif; ?>

        <!-- Form -->
        <form action="addFacility.php" method="post" class="p-4 bg-light rounded shadow-sm mx-auto" style="max-width: 800px;">

            <!-- Title and Category -->
            <div class="form-row">
                <div class="form-group col-md-6">
                    <label for="title">Title:</label>
                    <input type="text" id="title" name="title" class="form-control"
                           value="<?= isset($_POST['title']) ? htmlspecialchars($_POST['title'], flags: ENT_QUOTES) : ''; ?>"
                           placeholder="Enter facility title" required>
                </div>
                <div class="form-group col-md-6">
                    <label for="category">Category:</label>
                    <input type="text" id="category" name="category" class="form-control"
                           value="<?= isset($_POST['category']) ? htmlspecialchars($_POST['category'], flags: ENT_QUOTES) : ''; ?>"
                           placeholder="Enter facility category" required>
                </div>
            </div>

            <!-- Description -->
            <div class="form-group">
                <label for="description">Description:</label>
                <textarea id="description" name="description" rows="4" class="form-control"
                          placeholder="Enter a brief description of the facility" required><?= isset($_POST['description']) ? htmlspecialchars($_POST['description'], flags: ENT_QUOTES) : ''; ?></textarea>
            </div>

            <!-- Address (House Number, Street, Town, County, Postcode) -->
            <div class="form-row">
                <div class="form-group col-md-4">
                    <label for="houseNumber">House Number:</label>
                    <input type="text" id="houseNumber" name="houseNumber" class="form-control"
                           value="<?= isset($_POST['houseNumber']) ? htmlspecialchars($_POST['houseNumber'], flags: ENT_QUOTES) : ''; ?>"
                           placeholder="House No." required>
                </div>
                <div class="form-group col-md-8">
                    <label for="streetName">Street Name:</label>
                    <input type="text" id="streetName" name="streetName" class="form-control"
                           value="<?= isset($_POST['streetName']) ? htmlspecialchars($_POST['streetName'], flags: ENT_QUOTES) : ''; ?>"
                           placeholder="Street Name" required>
                </div>
            </div>

            <div class="form-row">
                <div class="form-group col-md-4">
                    <label for="county">County:</label>
                    <input type="text" id="county" name="county" class="form-control"
                           value="<?= isset($_POST['county']) ? htmlspecialchars($_POST['county'], flags: ENT_QUOTES) : ''; ?>"
                           placeholder="County" required>
                </div>
                <div class="form-group col-md-4">
                    <label for="town">Town:</label>
                    <input type="text" id="town" name="town" class="form-control"
                           value="<?= isset($_POST['town']) ? htmlspecialchars($_POST['town'], flags: ENT_QUOTES) : ''; ?>"
                           placeholder="Town" required>
                </div>
                <div class="form-group col-md-4">
                    <label for="postcode">Postcode:</label>
                    <input type="text" id="postcode" name="postcode" class="form-control"
                           value="<?= isset($_POST['postcode']) ? htmlspecialchars($_POST['postcode'], flags: ENT_QUOTES) : ''; ?>"
                           placeholder="Postcode" required>
                </div>
            </div>
```

```
            <!--Status -->
            <div class="form-group col-md-8">
                <label for="status">Status:</label>
                <input type="text" id="streetName" name="status" class="form-control"
                       value="<?= isset($_POST['status']) ? htmlspecialchars($_POST['status'], flags: ENT_QUOTES) : ''; ?>"
                       placeholder="Status" required>
            </div>

            <!-- Coordinates (Longitude and Latitude) -->
            <div class="form-row">
                <div class="form-group col-md-6">
                    <label for="longitude">Longitude:</label>
                    <input type="text" id="longitude" name="longitude" class="form-control"
                           value="<?= isset($_POST['longitude']) ? htmlspecialchars($_POST['longitude'], flags: ENT_QUOTES) : ''; ?>"
                           placeholder="Longitude" required>
                </div>

                <div class="form-group col-md-6">
                    <label for="latitude">Latitude:</label>
                    <input type="text" id="latitude" name="latitude" class="form-control"
                           value="<?= isset($_POST['latitude']) ? htmlspecialchars($_POST['latitude'], flags: ENT_QUOTES) : ''; ?>"
                           placeholder="Latitude" required>
                </div>



            </div>


            </div>

            <!-- Buttons -->
            <div class="form-group text-right">
                <button type="submit" class="btn btn-primary">Add Facility</button>
                <a href="index.php" class="btn btn-secondary">Cancel</a>
            </div>

        </form>
    </div>
```

```
        </form>
    </div>

<?php require('template/footer.phtml'); ?>
```

ecoFacilityDetails.phtml

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title><?= htmlspecialchars($facility->getTitle()) ?> - Facility Details</title>
    <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css">
    <link href="/css/my-style.css" rel="stylesheet">
</head>

<body>
<!-- Go back to the previous page -->
<button onclick="history.back()" style="margin: 20px;">Back</button>

<div class="container mt-5">
    <h1><?= htmlspecialchars($facility->getTitle()) ?></h1>

    <!-- Display the details of the selected ecoFacility -->
    <div class="card p-4">
        <p><strong>Category:</strong> <?= htmlspecialchars($facility->getCategory()) ?></p>
        <p><strong>Description:</strong> <?= htmlspecialchars($facility->getDescription()) ?></p>
        <p><strong>Address:</strong> <?= htmlspecialchars( string: $facility->getHouseNumber() . ' ' . $facility->getStreetName() . ', ' . $facility->getTown() . ', ' . $facility->getCounty() . ' - ' . $facility->getPostcod
        <p><strong>Location:</strong> Longitude <?= htmlspecialchars($facility->getLongitude()) ?>, Latitude <?= htmlspecialchars($facility->getLatitude()) ?></p>
        <p><strong>Contributor:</strong> <?= htmlspecialchars($facility->getContributor()) ?></p>
        <p><strong>Status:</strong> <?= htmlspecialchars($facility->getStatus()) ?></p>

        <!-- Display the Edit button for admins -->
        <?php if (isset($_SESSION['userType']) && $_SESSION['userType'] == 1): ?>
            <div class="mt-3">
                <a href="editFacility.php?id=<?= htmlspecialchars($facility->getID()); ?>" class="btn btn-warning edit-button">Edit</a>
            </div>
        <?php endif; ?>

    </div>



</div>
</body>
</html>
```

editFacility.phtml

```php
<?php require('template/header.phtml'); ?>
<!-- Include the header template -->

<h3>Edit Eco-Facility</h3>

<!-- Display error message if any -->
<?php if (isset($view->errorMessage) && !empty($view->errorMessage)): ?>
    <div class="alert alert-danger">
        <?= htmlspecialchars($view->errorMessage, flags: ENT_QUOTES); ?>
    </div>
<?php endif; ?>

<form method="post" action="editFacility.php?id=<?= htmlspecialchars($facility->getID(), flags: ENT_QUOTES); ?>">
    <!-- Hidden field to include the facility ID -->
    <input type="hidden" name="id" value="<?= htmlspecialchars($facility->getID(), flags: ENT_QUOTES); ?>">

    <!-- Input fields for each facility attribute -->
    <label for="title">Title:</label>
    <input
            type="text"
            id="title"
            name="title"
            value="<?= htmlspecialchars($facility->getTitle(), flags: ENT_QUOTES); ?>"
            required
    >
    <br><br>

    <label for="category">Category:</label>
    <input
            type="text"
            id="category"
            name="category"
            value="<?= htmlspecialchars($facility->getCategory(), flags: ENT_QUOTES); ?>"
            required
    >
    <br><br>

    <label for="description">Description:</label>
    <textarea
            id="description"
            name="description"
```

```
            required
        ><?= htmlspecialchars($facility->getDescription(), flags: ENT_QUOTES); ?></textarea>
        <br><br>

        <label for="houseNumber">House Number:</label>
        <input
                type="text"
                id="houseNumber"
                name="houseNumber"
                value="<?= htmlspecialchars($facility->getHouseNumber(), flags: ENT_QUOTES); ?>"
                required
        >
        <br><br>

        <label for="streetName">Street Name:</label>
        <input
                type="text"
                id="streetName"
                name="streetName"
                value="<?= htmlspecialchars($facility->getStreetName(), flags: ENT_QUOTES); ?>"
                required
        >
        <br><br>

        <label for="county">County:</label>
        <input
                type="text"
                id="county"
                name="county"
                value="<?= htmlspecialchars($facility->getCounty(), flags: ENT_QUOTES); ?>"
                required
        >
        <br><br>

        <label for="town">Town:</label>
        <input
                type="text"
                id="town"
                name="town"
```

```
            value="<?= htmlspecialchars($facility->getTown(), flags: ENT_QUOTES); ?>"
            required
    >
    <br><br>

    <label for="postcode">Postcode:</label>
    <input
            type="text"
            id="postcode"
            name="postcode"
            value="<?= htmlspecialchars($facility->getPostcode(), flags: ENT_QUOTES); ?>"
            required
    >
    <br><br>

    <label for="longitude">Longitude:</label>
    <input
            type="text"
            id="longitude"
            name="longitude"
            value="<?= htmlspecialchars($facility->getLongitude(), flags: ENT_QUOTES); ?>"
            required
    >
    <br><br>

    <label for="latitude">Latitude:</label>
    <input
            type="text"
            id="latitude"
            name="latitude"
            value="<?= htmlspecialchars($facility->getLatitude(), flags: ENT_QUOTES); ?>"
            required
    >

    <br><br>

    <label for="status">Status:</label>
    <input
            type="text"
```

```
            id="status"
            name="status"
            value="<?= htmlspecialchars($facility->getStatus(), flags: ENT_QUOTES); ?>"
            required
    >

    <br><br>

    <!-- Submit Button -->
    <button type="submit" name="updateFacility">Update Facility</button>
    <button type="button" onclick="window.location.href='index.php';">Cancel</button>
</form>
```

index.phtml

```php
<?php require('template/header.phtml') ?>

    <h3>Welcome to ecoBuddy</h3>

    <!-- Check if a user is logged in and is an admin-->
<?php if (isset($_SESSION['userType']) && $_SESSION['userType'] == 1): ?>
    <!-- Add the Facility button for Admin -->
    <a href="addFacility.php" class="btn btn-success mb-3">Add New Facility</a>
<?php endif; ?>

    <!-- Search Form -->
    <h2>Search for an Eco-Facility</h2>
    <form method="get" action="index.php">
        <div class="form-group">
            <input type="text" id="liveSearchInput" name="query" placeholder="Search facilities..." autocomplete="off"
                   value="<?= htmlspecialchars( string: $_GET['query'] ?? '') ?>" class="form-control w-25 d-inline">

            <!-- Container for Live Search Results -->
            <div id="searchResults" class="list-group w-50 position-absolute bg-white"></div>

            <!-- Category dropdown-->
            <select name="category" class="form-control w-25 d-inline">
                <option value="">All Categories</option>
                <option value="1" <?= ($_GET['category'] ?? '') == '1' ? 'selected' : '' ?>>1</option>
                <option value="2" <?= ($_GET['category'] ?? '') == '2' ? 'selected' : '' ?>>2</option>
                <option value="3" <?= ($_GET['category'] ?? '') == '3' ? 'selected' : '' ?>>3</option>
                <option value="4" <?= ($_GET['category'] ?? '') == '3' ? 'selected' : '' ?>>4</option>
                <option value="5" <?= ($_GET['category'] ?? '') == '3' ? 'selected' : '' ?>>5</option>
                <option value="6" <?= ($_GET['category'] ?? '') == '3' ? 'selected' : '' ?>>6</option>
                <option value="7" <?= ($_GET['category'] ?? '') == '3' ? 'selected' : '' ?>>7</option>
                <option value="8" <?= ($_GET['category'] ?? '') == '3' ? 'selected' : '' ?>>8</option>
                <option value="9" <?= ($_GET['category'] ?? '') == '3' ? 'selected' : '' ?>>9</option>
                <option value="10" <?= ($_GET['category'] ?? '') == '3' ? 'selected' : '' ?>>10</option>
                <option value="11" <?= ($_GET['category'] ?? '') == '3' ? 'selected' : '' ?>>11</option>
                <option value="12" <?= ($_GET['category'] ?? '') == '3' ? 'selected' : '' ?>>12</option>
                <option value="13" <?= ($_GET['category'] ?? '') == '3' ? 'selected' : '' ?>>13</option>

            </select>
            <!--- Submit search button-->
            <button type="submit" class="btn btn-primary">Search</button>
        </div>
```

```
    </form>

    <!-- Display DB Messages if needed -->
<?php if (!empty($view->dbMessage)): ?>
    <p class="text-danger"><?= htmlspecialchars($view->dbMessage) ?></p>
<?php endif; ?>

    <!-- Leaflet Map : CSS/JS -->
    <link rel="stylesheet" href="https://unpkg.com/leaflet@1.9.4/dist/leaflet.css"/>
    <script src="https://unpkg.com/leaflet@1.9.4/dist/leaflet.js"></script>

    <!-- Map Container -->
    <div id="mapDiv" style="..."></div>

    <!-- Facilities Table -->
    <table class="table table-bordered">
        <thead>
        <tr>
            <th>ID</th>
            <th>Title</th>
            <th>Category</th>
            <th>Description</th>
            <th>Address</th>
            <th>Location</th>
            <th>Contributor</th>
            <th>Status</th>
            <!-- Admin Actions -->
            <?php if (isset($_SESSION['userType']) && $_SESSION['userType'] == 1): ?>
                <th>Actions</th>
            <?php endif; ?>
        </tr>
        </thead>
        <tbody>
        <?php foreach ($view->ecoFacilities as $ecoFacilityData): ?>
            <tr>
                <td><?= htmlspecialchars($ecoFacilityData->getID()); ?></td>
                <td><?= htmlspecialchars($ecoFacilityData->getTitle()); ?></td>
                <td><?= htmlspecialchars($ecoFacilityData->getCategory()); ?></td>
                <td><?= htmlspecialchars($ecoFacilityData->getDescription()); ?></td>
                <td>
                    <?= htmlspecialchars( string: $ecoFacilityData->getHouseNumber() . " " . $ecoFacilityData->getStreetName() . ", " .
```

```
                    $ecoFacilityData->getTown() . ", " . $ecoFacilityData->getCounty() . " - " . $ecoFacilityData->getPostcode()); ?>
                </td>
                <td>
                    <?= "Longitude: " . htmlspecialchars($ecoFacilityData->getLongitude()) . ", Latitude: " . htmlspecialchars($ecoFacilityData->getLatitude()); ?>
                </td>
                <td><?= htmlspecialchars($ecoFacilityData->getContributor()); ?></td>
                <td><?= htmlspecialchars($ecoFacilityData->getStatus()); ?></td>
                <!-- Admin Actions -->
                <?php if (isset($_SESSION['userType']) && $_SESSION['userType'] == 1): ?>
                    <td>
                        <a href="editFacility.php?id=<?= htmlspecialchars($ecoFacilityData->getID()); ?>" class="btn btn-warning btn-sm">Edit</a>
                        <a href="deleteFacility.php?id=<?= htmlspecialchars($ecoFacilityData->getID()); ?>" class="btn btn-danger btn-sm">Delete</a>
                    </td>
                <?php endif; ?>
            </tr>
        <?php endforeach; ?>
        </tbody>
    </table>

    <!-- Pagination Controls -->
    <div>
        <p>Page <?= $view->currentPage ?> of <?= $view->totalPages ?></p>
        <nav>
            <ul class="pagination">
                <!-- Previous page Link -->
                <?php if ($view->currentPage > 1): ?>
                    <li class="page-item">
                        <a class="page-link" href="?page=<?= $view->currentPage - 1 ?>&query=<?= urlencode( string: $_GET['query'] ?? '') ?>&category=<?= urlencode( string: $_GET['category'] ?? '') ?>">
                            Previous
                        </a>
                    </li>
                <?php endif; ?>

                <!-- Page number links -->
                <?php for ($i = 1; $i <= $view->totalPages; $i++): ?>
                    <li class="page-item <?= $i == $view->currentPage ? 'active' : '' ?>">
                        <a class="page-link" href="?page=<?= $i ?>&query=<?= urlencode( string: $_GET['query'] ?? '') ?>&category=<?= urlencode( string: $_GET['category'] ?? '') ?>">
                            <?= $i ?>
```

```
                    </a>
                </li>
            <?php endfor; ?>

            <!-- Next Page Link -->
            <?php if ($view->currentPage < $view->totalPages): ?>
                <li class="page-item">
                    <a class="page-link" href="?page=<?= $view->currentPage + 1 ?>&query=<?= urlencode( string: $_GET['query'] ?? '') ?>&category=<?= urlencode( string: $_GET['category'] ?? '') ?>">
                        Next
                    </a>
                </li>
            <?php endif; ?>
        </ul>
    </nav>
</div>

<!-- map.js -->
<script src="js/map.js"></script>

<!--liveSearch.js -->
<script src="js/liveSearch.js"></script>

<!-- Live seracch script  -->
<script>
    document.getElementById('liveSearchInput').addEventListener('input', function() {
        const query = this.value.trim();
        //Hide results box if input is empty
        if (query === '') {
            document.getElementById('searchResults').innerHTML = '';
            document.getElementById('searchResults').style.display = 'none';
            return;
        }

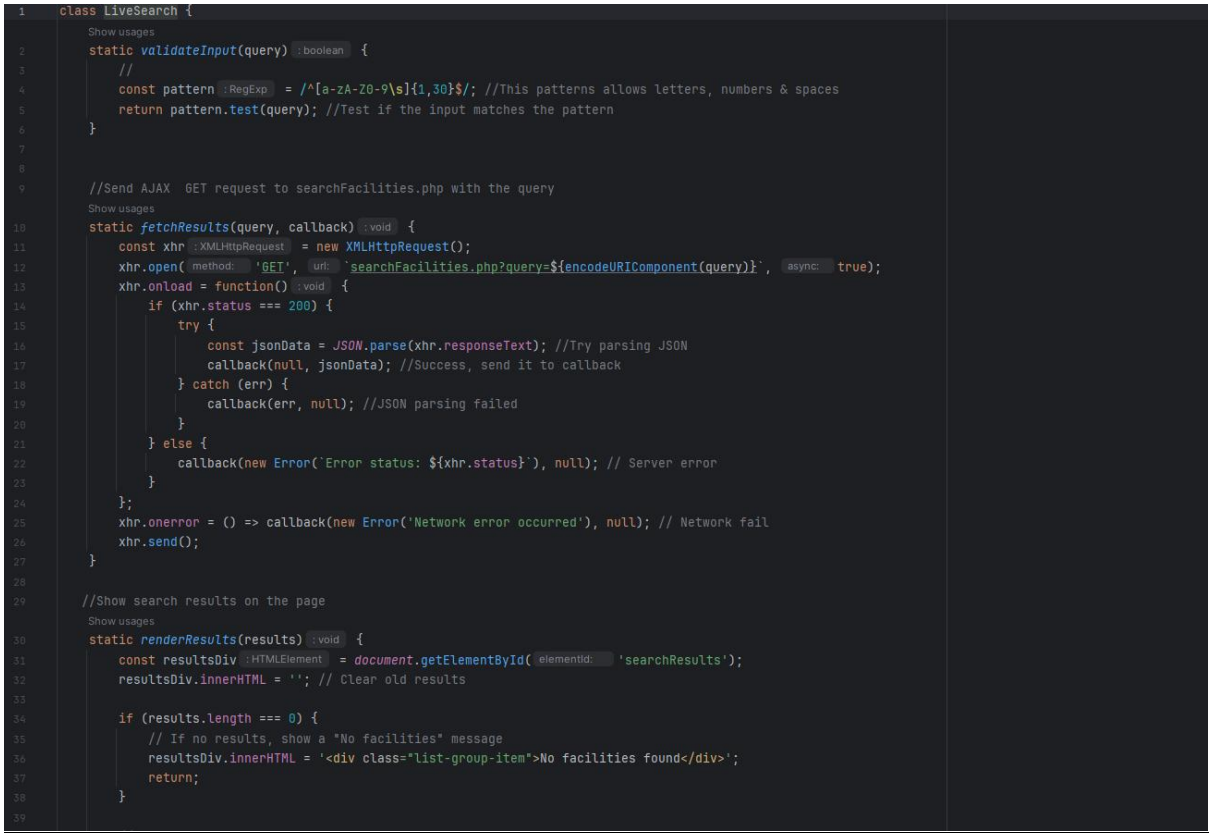
        //If input is valid, fetch and show live results
        if (LiveSearch.validateInput(query)) {
            LiveSearch.fetchResults(query, function(error, data) {
                if (error) {
```

```
                    console.error("Fetching error:", error);
                    document.getElementById('searchResults').innerHTML = '<div class="list-group-item text-danger">Error loading results</div>';
                    document.getElementById('searchResults').style.display = 'block';
                    return;
                }
                LiveSearch.renderResults(data);
                document.getElementById('searchResults').style.display = 'block';
            });
        } else {
            //Show validation error message
            document.getElementById('searchResults').innerHTML = '<div class="list-group-item text-warning">Invalid input detected.</div>';
            document.getElementById('searchResults').style.display = 'block';
        }
    });
</script>

<?php require('template/footer.phtml') ?>
```

JS Files

liveSearch.js

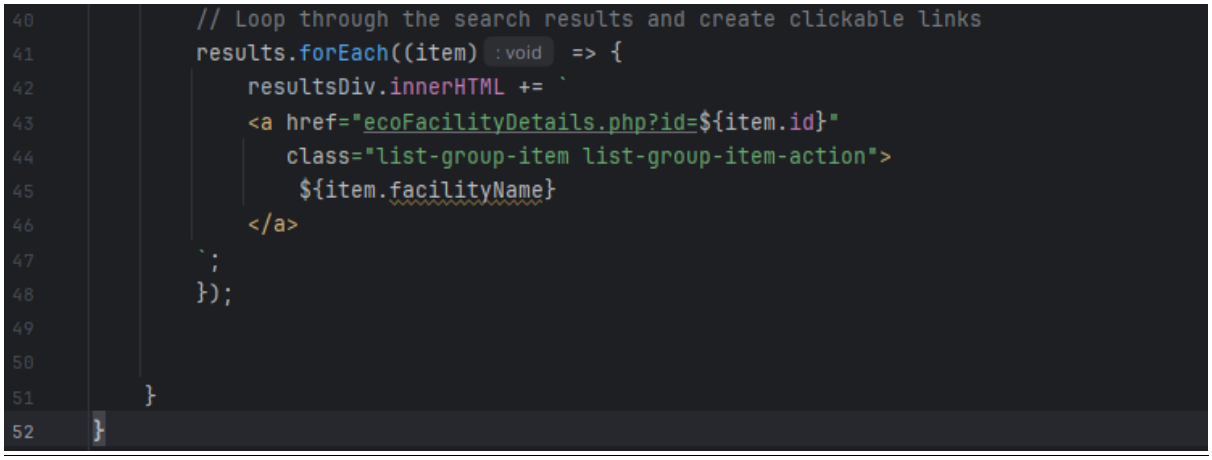


```
class LiveSearch {
    Show usages
    static validateInput(query) :boolean {
        //
        const pattern :RegExp = /^[a-zA-Z0-9\s]{1,30}$/; //This patterns allows letters, numbers & spaces
        return pattern.test(query); //Test if the input matches the pattern
    }


    //Send AJAX  GET request to searchFacilities.php with the query
    Show usages
    static fetchResults(query, callback) :void {
        const xhr :XMLHttpRequest = new XMLHttpRequest();
        xhr.open( method: 'GET', url: `searchFacilities.php?query=${encodeURIComponent(query)}`, async: true);
        xhr.onload = function() :void {
            if (xhr.status === 200) {
                try {
                    const jsonData = JSON.parse(xhr.responseText); //Try parsing JSON
                    callback(null, jsonData); //Success, send it to callback
                } catch (err) {
                    callback(err, null); //JSON parsing failed
                }
            } else {
                callback(new Error(`Error status: ${xhr.status}`), null); // Server error
            }
        };
        xhr.onerror = () => callback(new Error('Network error occurred'), null); // Network fail
        xhr.send();
    }

    //Show search results on the page
    Show usages
    static renderResults(results) :void {
        const resultsDiv :HTMLElement = document.getElementById( elementId: 'searchResults');
        resultsDiv.innerHTML = ''; // Clear old results

        if (results.length === 0) {
            // If no results, show a "No facilities" message
            resultsDiv.innerHTML = '<div class="list-group-item">No facilities found</div>';
            return;
        }
```



```
            // Loop through the search results and create clickable links
            results.forEach((item) :void => {
                resultsDiv.innerHTML += `
                <a href="ecoFacilityDetails.php?id=${item.id}"
                   class="list-group-item list-group-item-action">
                    ${item.facilityName}
                </a>
            `;
            });


        }
    }
```

map.js

```
const map = L.map('mapDiv').setView([53.4808, -2.2426], 13);  //Initialize the map on Manchester

// Initialize OpenStreetMap tiles for the map's background
L.tileLayer('https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png', {
    attribution: '&copy; OpenStreetMap contributors'
}).addTo(map);

// Fetch facilities from backend usin AJAX
Show usages
async function fetchFacilities() : Promise<any> {
    const response : Response  = await fetch( input: 'fetchFacilities.php');
    return response.json();
}

// Validate latitude & longitude
Show usages
function isValidLatLng(lat, lng){
    return isFinite(lat) && isFinite(lng) && Math.abs(lat) <= 90 && Math.abs(lng) <= 180;
}

// Add markers dynamically to map
Show usages
async function displayFacilitiesOnMap() : Promise<void> {
    const facilities = await fetchFacilities();

    facilities.forEach(facility => {
        // Check if the latitude and longitude are valid
        if(isValidLatLng(facility.latitude, facility.longitude)){
            //Add a marker to the map at the facility's coordinates with a popup too when clicked
            L.marker([facility.latitude, facility.longitude])
                .addTo(map)
                .bindPopup(`📍<b>${facility.name}</b>`);
        }
    });
}

// Current user's location
map.locate({setView: true, maxZoom: 15});

// Handle the error if the user's location cannot be determined
map.on('locationerror', (e) : void  => alert(`Could not get your location: ${e.message}`));
```

```
// When the user's location is found, add a marker at their location with a popup saying "📍Your location!"
map.on('locationfound', (e) => L.marker(e.latlng).addTo(map).bindPopup('📍Your location!'));

// Initialize
displayFacilitiesOnMap();
```

CSS Files

my-style.css

```css
#title {
    margin-top: 12px;
    background-color: #fff;
    color: #000;

}
#menu {
    border-top: solid 6px #000;
    background-color: #fff;
    color: #fff;
    height: 400px;
}

#menu a {
    background-color: #884242;
    color: #fff;
    text-decoration: none;
    display: block;
}
#menu a:hover {
    background-color: #f00;
    color: #ddd;
    text-decoration:underline;
    display: block;
}

#content {
    background-color: #fff;
    border-top: solid 6px #f00;

}

#footer {
    margin_top: 20px;
    text-align: center;
    background-color: #000;
    color: #fff;
}

/* Apply to all tables */
table {
    width: 100%;
```

```
    border-collapse: collapse;
    margin: 20px 0;
    font-size: 14px;
}

table th, table td {
    border: 1px solid #ddd;
    padding: 8px;
    text-align: left;
}

table th {
    background-color: #f4f4f4;
    color: #333;
}

table tr:nth-child(even) {
    background-color: #f9f9f9;
}

table tr:hover {
    background-color: #f1f1f1;
}

.scrollable-dropdown {
    max-height: 150px; /* Set the maximum height */
    overflow-y: auto;  /* Enable vertical scrolling */
}


#searchResults {
    max-height: 300px; /* Limit dropdown height */
    overflow-y: auto; /* Enable scrolling */
    border: 1px solid #ddd;
    border-radius: 5px;
    box-shadow: 0 2px 5px rgba(0,0,0,0.2);
    z-index: 1000;
}

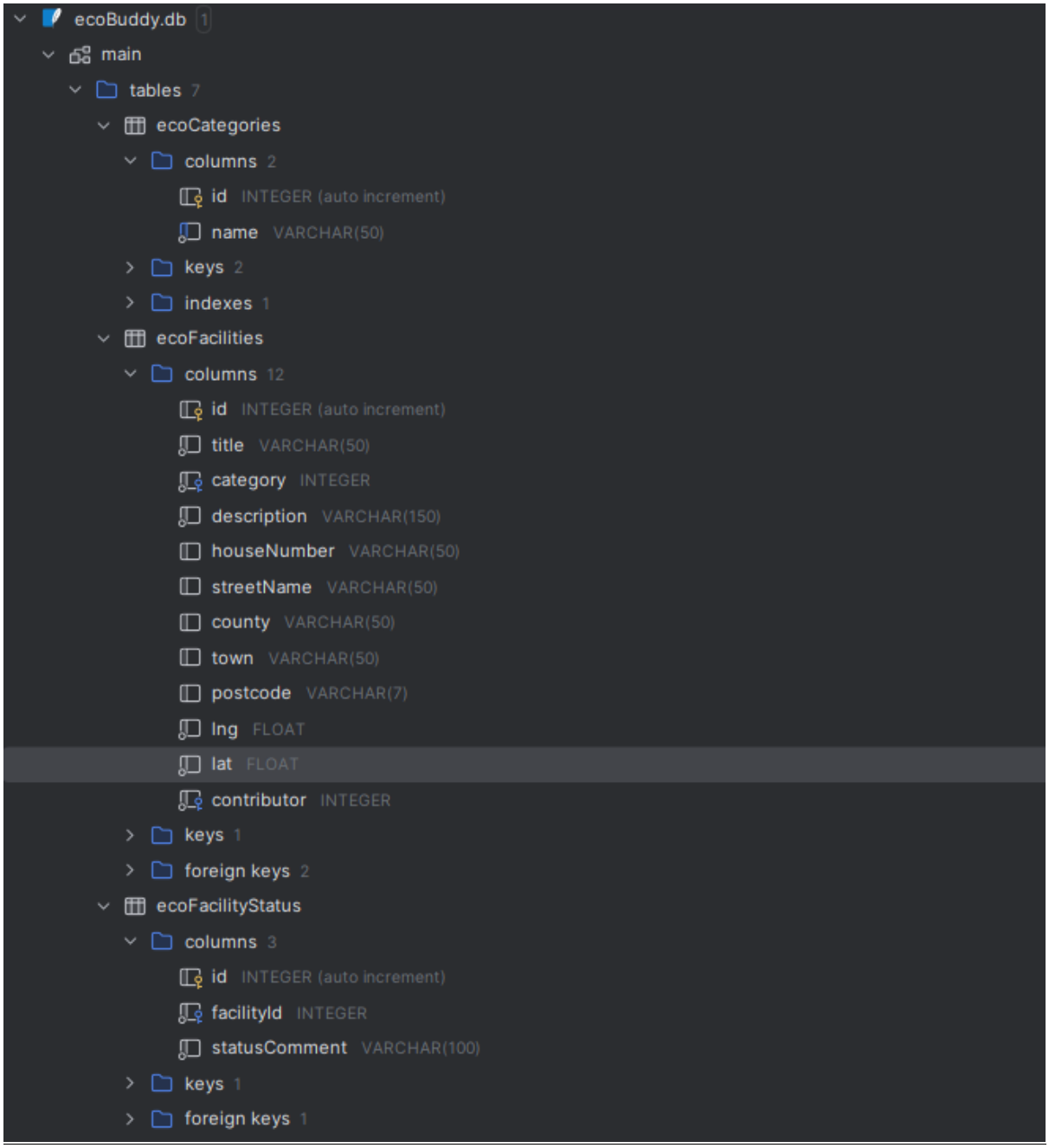
#searchResults .list-group-item {
    padding: 8px 16px;
```

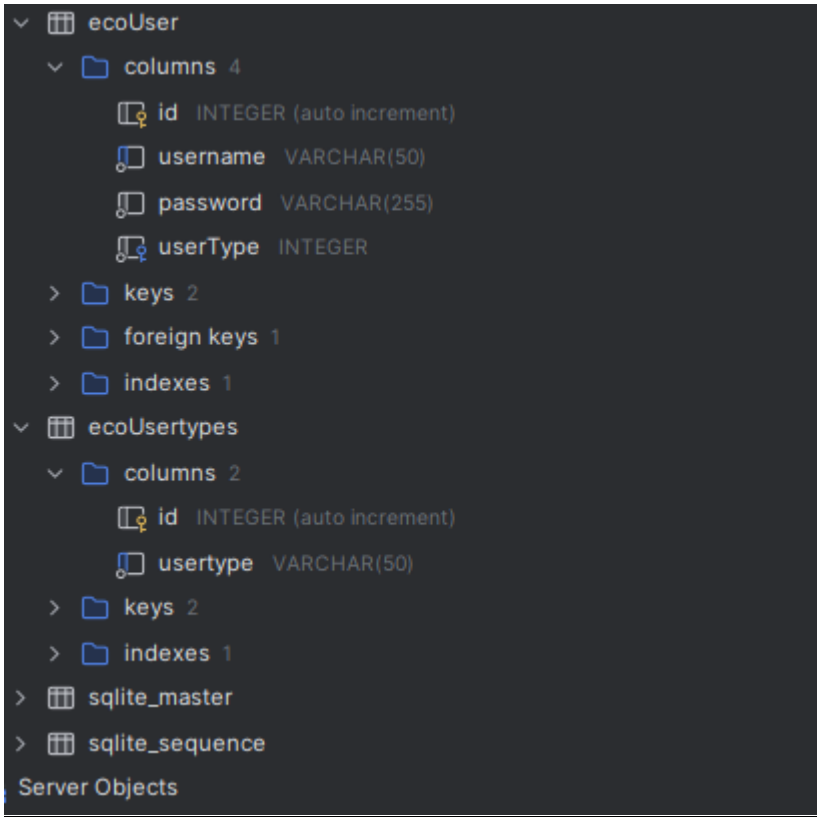
```
    font-size: 14px; /* Compact font size */
    white-space: nowrap; /* Prevent text wrapping */
    overflow: hidden; /* Hide overflow */
    text-overflow: ellipsis; /* Add an ellipsis for long text */
}

#searchResults .list-group-item:hover {
    background-color: #f8f9fa; /* Highlight hover */
}


.card {
    background-color: #ffffff;
    border: 1px solid #ddd;
    border-radius: 10px;
    padding: 20px;
    margin-bottom: 20px;
    box-shadow: 0 2px 8px rgba(0, 0, 0, 0.05);
}
.edit-button {
    position: absolute;
    top: 15px;
    right: 15px;
    padding: 10px 20px;
    font-size: 18px;
    border-radius: 8px;
    background-color: #000;
    color: #fff;
    text-decoration: none;
}
```

Database Records (also database schema)


ecoBuddy.db 1
main
tables 7
ecoCategories
columns 2
id INTEGER (auto increment)
name VARCHAR(50)
keys 2
indexes 1
ecoFacilities
columns 12
id INTEGER (auto increment)
title VARCHAR(50)
category INTEGER
description VARCHAR(150)
houseNumber VARCHAR(50)
streetName VARCHAR(50)
county VARCHAR(50)
town VARCHAR(50)
postcode VARCHAR(7)
lng FLOAT
lat FLOAT
contributor INTEGER
keys 1
foreign keys 2
ecoFacilityStatus
columns 3
id INTEGER (auto increment)
facilityId INTEGER
statusComment VARCHAR(100)
keys 1
foreign keys 1



ecoFacilities table (there's 100+ records)

	id	title	category	description	houseNumber	streetName	county	town	postcode	lng	lat	contributor	stati
	ilter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	1	Facility 19	1	Description for facility 1	12	Main St	County A	Town X	AB12CD	-2.234567	53.56789	1	NULL
2	2	Facility 2	2	Description for facility 2	48	Second St	County B	Town Y	CD34EF	-1.987654	52.345678	2	NULL
3	3	Facility 3	3	Description for facility 3	78	Park Ave	County C	Town Z	EF56GH	-3.123456	51.234567	3	NULL
4	4	Facility 4	1	Description for facility 4	56	Elm St	County A	Town X	GH78IJ	-2.345678	54.123456	1	NULL
5	5	Facility 5	2	Description for facility 5	23	High St	County D	Town W	IJ90KL	-1.456789	50.987654	2	NULL
6	6	Facility 6	3	Description for facility 6	89	Broadway	County C	Town Y	KL12MN	-4.56789	49.876543	3	NULL
7	7	Facility 7	1	Description for facility 7	34	Maple Rd	County B	Town W	MN34OP	-5.678901	48.765432	1	NULL
8	8	Facility 8	2	Description for facility 8	67	Oak Ln	County D	Town Z	OP56QR	-6.789012	47.654321	2	NULL
9	9	Facility 9	3	Description for facility 9	12A	Ash Ave	County A	Town X	QR78ST	-7.890123	46.54321	3	NULL
10	11	Facility 10	1	Description for facility 10	24	Kobe Way	LA	Manchester	M4 6LY	2.123456	52.234567	admin	NULL
11	17	Arndale	1	arndale	1	Arndale	Lala Land	Salford	M5 5HL	-2.287306	53.489105	admin	NULL
12	19	Idk	Recycling	Who knows	11	WhoKnows	Lala Land	Salford	AB12CD	-2.24189	53.48333	admin	NULL
13	20	Castlefield	1	Letsgo	1	Castlefield	County A	Town X	M5 5HL	-2.248729	53.472345	admin	NULL
14	21	Whitworth	Recycling	Lol	132	Whitworth	County A	Salford	M6 5ID	-2.235417	53.474958	admin	NULL
15	22	RealDale	Recycling	ok	134	Arndale	Lala Land	Manchester	M5 5HL	-2.24139	53.48333	admin	NULL
16	23	Barrett Group	3	Issue decision hold compare. Gas bas…	510	Chapman Field	Greater Manchester	Manchester	52837	-2.251117	53.488302	3	pendi
17	24	Greenworks Group	3	Local road expert event sound …	92	Broadway Road	Greater Manchester	Manchester	M143ND	-2.248712	53.474001	2	activ
18	25	Eco Logic Ltd.	5	Market eye night similar space beat …	14	Victoria Avenue	Greater Manchester	Manchester	M204RH	-2.251774	53.49211	1	inact
19	26	SunCore Solutions	8	Surface series during school low goa…	77	King Street	Greater Manchester	Manchester	M112QW	-2.23988	53.470012	3	pendi
20	27	Greenworks Group	3	Local road expert event sound …	92	Broadway Road	Greater Manchester	Manchester	M143ND	-2.248712	53.474001	2	activ
21	28	Eco Logic Ltd.	5	Market eye night similar space beat …	14	Victoria Avenue	Greater Manchester	Manchester	M204RH	-2.251774	53.49211	1	inact
22	29	SunCore Solutions	8	Surface series during school low goa…	77	King Street	Greater Manchester	Manchester	M112QW	-2.23988	53.470012	3	pendi
23	30	Company 1	4	Research innovative solutions green …	156	Oxford Road	Greater Manchester	Manchester	M133HX	-2.30576	53.479623	3	activ

Table: ecoFacilities

Filter in any column

	id	title	category	description	houseNumber	streetName	county	town	postcode	lng	lat	contributor	sta
	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
35	42	GreenTech Innovators	13	A cutting-edge technology hub …	72	King Street	Greater Manchester	Manchester	M153WA	-2.228667	53.472134	1	acti
36	43	EcoShift Solutions	8	Bringing businesses and communities …	45	Crescent Road	Greater Manchester	Manchester	M208JS	-2.230156	53.478227	2	inac
37	44	Urban Harvest	9	A community garden initiative growin…	12	Station Road	Greater Manchester	Manchester	M100EN	-2.238926	53.474548	3	pend
38	45	Recycling Revolution	4	A state-of-the-art facility for …	88	Union Street	Greater Manchester	Manchester	M145DP	-2.24711	53.463989	1	acti
39	46	Compost City	5	Eco-friendly composting services for…	29	Victoria Road	Greater Manchester	Manchester	M121PB	-2.240781	53.470915	2	inac
40	47	The Green Building	13	A model building showcasing energy-…	77	Chapel Street	Greater Manchester	Manchester	M35RF	-2.23352	53.4758	3	pend
41	48	Solar Solutions	8	A company providing solar panel …	15	Granby Street	Greater Manchester	Manchester	M220QW	-2.24218	53.470801	1	acti
42	49	EcoGro	6	An urban farm that grows sustainable…	54	Newcastle Road	Greater Manchester	Manchester	M141DA	-2.23832	53.476473	2	inac
43	50	Nature's Path	7	An eco-friendly path through the cit…	99	Oxford Street	Greater Manchester	Manchester	M128HR	-2.226702	53.469555	3	acti
44	51	Clean Earth Solutions	3	A consultancy firm offering …	56	Deansgate	Greater Manchester	Manchester	M146NY	-2.24682	53.477234	1	inac
45	52	ReNew Energy	8	A renewable energy company …	89	Cross Street	Greater Manchester	Manchester	M109AU	-2.242684	53.470671	2	pend
46	53	BioRevolution	4	A recycling center that specializes …	45	Holly Street	Greater Manchester	Manchester	M138QF	-2.231238	53.465102	3	acti
47	54	Sustain Your City	7	A green initiative aiming to make th…	23	King Street	Greater Manchester	Manchester	M159DZ	-2.232349	53.472967	1	inac
48	55	Eco Living Collective	12	An eco-friendly lifestyle store …	34	Liverpool Road	Greater Manchester	Manchester	M144DQ	-2.234879	53.468021	2	pend
49	56	The Green Space	6	A community center providing …	100	Wilmslow Road	Greater Manchester	Manchester	M145JH	-2.244895	53.470299	3	acti
50	57	Eco Conscious Living	9	An urban space for people to learn …	22	Albert Road	Greater Manchester	Manchester	M153NJ	-2.229491	53.47366	1	inac
51	58	Urban Bloom	6	A community initiative dedicated to …	12	Pine Street	Greater Manchester	Manchester	M192EA	-2.24383	53.466899	1	acti
52	59	SolarTech Ventures	13	Providing solar energy solutions for…	67	Lombard Street	Greater Manchester	Manchester	M276RB	-2.24042	53.47255	2	pend
53	60	EcoCity Solutions	8	Innovative solutions for sustainable…	34	Roseville Road	Greater Manchester	Manchester	M187LR	-2.230803	53.463892	3	inac
54	61	The Green Store	12	Eco-friendly products for a …	14	Eastgate Street	Greater Manchester	Manchester	M192DZ	-2.24157	53.47501	1	acti
55	62	Sustainability Co-op	9	A community-driven initiative …	99	Oxford Road	Greater Manchester	Manchester	M142QZ	-2.227017	53.469158	2	inac
56	63	Eco Metro	13	An energy-efficient public …	51	Hightown Road	Greater Manchester	Manchester	M192EE	-2.238473	53.466173	3	acti
57	64	GreenPark Plaza	6	A sustainable office space offering …	87	Park Avenue	Greater Manchester	Manchester	M271SD	-2.234839	53.471618	1	inac

ecoUser (there's 1000+ records)

	id	username	password	userType
	Filter	Filter	Filter	Filter
4	6	admin	G7!fY8mZ$2k@PwR4	1
5	8	lee	H7k@M2!jXwY9r&Lp	2
6	34	oliver_white	P@ssw0rd1234!	1
7	35	amelia_james	S3cur3Passw0rd!@#	1
8	36	jackson_brown	Str0ngP@ssw0rd!5678	1
9	37	chloe_wilson	W1ns0rP@ssword2025!	1
10	38	luke_smith	P@ssw0rd!98J1d2	2
11	39	mia_miller	M1ll3rS3cure$2023	2
12	40	lucas_walker	L@ucasW4lk3r#45!	2
13	41	ella_harris	H@rrisP@ssw0rD22!	2
14	42	benjamin_carter	B3nj@minC4rt3r_!88	2
15	43	scarlett_lee	Sc@rL3tt_L33!89	2
16	44	henry_roberts	R0b3rts_H3nry2025$	2
17	45	sophia_evans	Soph!aEvan$2023!	2
18	46	daniel_perez	D@ni3l_P3r3z_123	2
19	47	lucy_jackson	L@cyJ@cks0n!2024	2
20	48	olivia_brown	Ol1v!@Br0wnP@ss	2
21	49	ryan_thomas	RyaN_T@h0m@5#2025	2
22	50	ella_martin	E!!aM@rtin1997*	2
23	51	james_wilson	J@mesW!ls0n2025	2
24	52	victoria_garcia	V!ct0r1@G@rC!A92	2
25	53	jack_morris	J@ckM0rr!s2025$	2
26	54	grace_davis	Gr@c3D@v!s_2019	2

ecoUsertypes

	id	usertype	name
	Fil...	Filter	Filter
1	1	1	Manager
2	2	2	User
3	3	3	Superuser

ecoFacilityStatus

	id	*facilityId*	statusComment
	Fil...	Filter	Filter
1	1	1	Operational and running efficiently.
2	2	2	Scheduled for maintenance next week.
3	3	3	Community events ongoing.
4	4	4	Closed temporarily for upgrades.

ecoCategories

	id	name
	Fil...	Filter
1	1	Transport
2	2	Waste Management
3	3	Shopping
4	4	Recycling Center
5	5	Composting Site
6	6	Eco Park
7	7	Nature Reserve
8	8	Renewable Energy Site
9	9	Community Garden
10	10	Wildlife Sanctuary
11	11	Organic Farm
12	12	Eco-Friendly Shop
13	13	Green Technology Hub